DATA STRUCTURE AND ALGORITHM

Programming Practice Session

31/10/2025

3rd Session

- Pointers (cont.)
- Stack
- Queue

Revision

```
int N = 1000; // 0x7ffe4
int *ptr1 = &N;
int *ptr2 = &N;
int *ptr3 = ptr1;
cout << ptr1 << " " << ptr2 << " " << ptr3;
*ptr1 = 100; cout << N;
*ptr2 = 200; cout << N;
*ptr3 = 300;
cout<< *ptr1 << " "<< *ptr2 << " "<< *ptr3 << " " << N;
```

Revision

```
int x = 10; // x address: 0x7fffc3
int *p = &x;  // p address: 0x7ff2c
int **pp = &p;
cout << x << endl;
cout << *p << endl;
cout << **pp << endl;
cout << &x << endl;
cout << p << endl;
cout << pp << endl;
```

Revision

```
int rows = 3, cols = 4;
    int **arr;
    // Cấp phát mảng 2 chiều bằng con trỏ kép
    arr = new int*[rows];
    for (int i = 0; i < rows; i++) {
        arr[i] = new int[cols];
    }
// Gán giá trị cho mảng
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            arr[i][j] = i * cols + j;
        }
    }
// In mảng ra màn hình
    cout << "Mảng 2D:\n";
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            cout << arr[i][j] << " ";
        }
        cout << endl;
    }

    // Giải phóng bộ nhớ
    for (int i = 0; i < rows; i++) {
        delete[] arr[i];
    }
    Delete [] arr;
```

Assignment 2

1. Implement a function to sort a dynamically allocated array in ascending order using the quicksort algorithm.
2. Find the maximum sum of a contiguous subarray within a dynamically allocated array.
3. Write a program to dynamically allocate a 2D array of size $m \times n$ (containing integers). Use a double pointer to allocate the 2D array dynamically, then:

- Calculate the sum of the elements on the main diagonal.
- Find the largest element located on the boundary of the matrix.
- Properly free all dynamically allocated memory.

4. Write functions to input and display a dynamically allocated matrix ($m \times n$). Then, implement a transpose function to convert rows into columns, storing the result in a new dynamically allocated matrix, and display the transposed matrix on the screen. Finally, free all dynamically allocated memory. (*Requirement: You must understand 2D dynamic memory allocation and pointer operations.*)

Assignment 2

5. Find the Next Greater Element for each element in an array.
For example, with the array a = `[4, 5, 2, 25]`:

The next greater element of 4 is 5.
The next greater element of 5 is 25.
The next greater element of 2 is 25.
The next greater element of 25 is -1 (no greater element).